WP2 · DATA ENGINEERING · FOR THE CTO & PLATFORM TEAM

From hospital systems to the 07:00 worklist

Finger Lakes Health System · 17 hospitals, Rochester, NY

FAST PATH · EVERY MORNING BY 07:00

Score ~275 new adult admissions at 06:30 and flag the top 30% (~77–82 patients/day) for Day-1 discharge planning.
Streams live admissions, ED diagnoses, and nursing notes—because public SPARCS arrives 60–180 days after discharge.

SLOW PATH · AFTER DISCHARGE

Track the monthly KPI (excess bed-days vs. the frozen 2017 P75 ruler) and retrain the model.
Combines the 08:00 case-manager write-back with final coded DRG billing (5–14 days after discharge).

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Data Analysis
UPM · November 2026

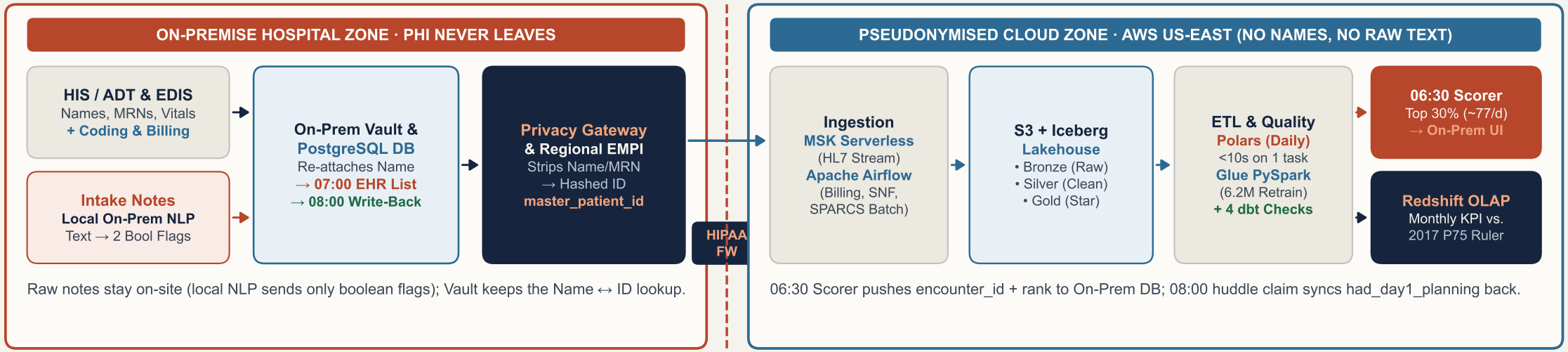
Four hospital systems and two public feeds replace a static file

Fills the 5 gaps in public SPARCS: patient ID, exact dates, diagnosis at admission, social factors, and nursing-home beds.

Source System & Owner	What It Records	Volume & Throughput (17 Hospitals)	Refresh Rate	Quality Issues to Handle
1. Admission & Bed (HIS/ADT) **Owner: Hospital IT**	Patient ID, age, sex, ZIP, payer, exact admit/discharge time, ward & bed.	137,867 stays/yr (~275 adult med-surg/d). ~15,000 HL7 msg/day (0.18 msg/s avg · 15–30 msg/s peak).	Real-time stream (HL7 ADT within seconds)	• Different local IDs per hospital. • Insurance often 'Pending' at night (updated next day).
2. Emergency Dept. (EDIS) **Owner: Emergency IT**	ED arrival flag ('Y'/'N'), vitals, and provisional admission diagnosis.	~600,000 ED visits/yr (~1,650/day; ~250/day become inpatient stays).	Real-time stream at admission order	• Admission diagnosis is rougher than final DRG (drifts in 15–20% of stays).
3. Nursing Intake Notes **Owner: Nursing Informatics**	Free-text Day-1 note: lives alone, family carer, stairs, prior SNF.	1–2 intake notes/stay (~500k/yr, ~15 MB/day of text).	Near real-time (within hours of reaching ward)	• Typos and negated text ('does not live alone'). • 05:00–06:00 notes may lag.
4. Coding & Billing **Owner: Finance & Coding**	Final APR-DRG, severity (1–4), total costs, and state SPARCS claim.	137,867 coded stays/yr (94,132 KPI-eligible = ~258/day).	Nightly batch (5–14 days after discharge)	• Not available on Day 1 (using it as a live input would be data leakage).
5. NY SNF Bed Feed **Owner: NYS DOH (uhyy-xp9s)**	Open beds at ~85 regional nursing homes accepting Medicare/Medicaid.	~85 facilities (~1,200 rows/update, <5 MB via REST API).	Daily batch at 06:00 AM	• Small rural nursing homes sometimes miss weekend updates.
6. Public SPARCS & Census **Owner: NYS DOH / US Census**	Historical stays (frozen 2017 P75 ruler) + ZIP deprivation score (1–10).	6,228,309 NY stays across 4 years (83 MB total in Parquet).	Yearly batch (download_sparcs.py)	• Caps LOS at '120 +', hides ~5,300 stays/yr, renames 3 columns after 2016.

HL7 peak (15–30 msg/s) occurs at 11:00–16:00 during ward discharges and ED handovers. 4-year SPARCS Parquet measured in ./data/ (83 MB).

Patient names and notes stay in the hospital; pseudonymised data scale in the cloud



Stage	Our Choice	Alternative Rejected	Why Our Choice Fits Better
Privacy Residency	Hybrid On-Premise + Cloud	All Cloud or All On-Prem	Keeps names & notes inside hospital firewall (HIPAA) while scaling analytics in AWS.
Streaming Broker	Amazon MSK Serverless (Kafka)	3-Node Kafka ($600/mo) / SQS	At 0.18 msg/s avg (30 msg/s peak), fixed brokers sit 99% idle; serverless costs ~$40/mo.
Orchestration	Apache Airflow	Linux cron scripts	Manages the 06:00 → 06:30 → 07:00 dependency chain with retries and failure alerts.
Storage & Tables	Amazon S3 + Apache Iceberg	HDFS or plain folders	$23/TB/mo storage with ACID MERGE updates when patients are discharged.
ETL & Scoring	Polars (daily) + Glue PySpark	24/7 Spark cluster	275 daily patients (<50 MB) score in <10s on 1 container ($0.02/run) vs. 3-min Spark boot.
Serving DBs	PostgreSQL (UI) + Redshift (BI)	Single database for both	Case managers need <15ms row lookups (OLTP); KPI queries scan 6.2M stays (OLAP).

Architecture adapts the course reference pipeline (Sessions 06–08) to HIPAA privacy rules and hospital volume.

Stream overnight admissions for the 07:00 cut-off; batch post-discharge coding

We do not stream everything—each source is ingested only as fast as the WP1 decision needs it.

REAL-TIME STREAMING · HL7 GATEWAY → MSK SERVERLESS

Sources: 1. HIS / ADT, 2. ED System, 3. Nursing Intake Flags

Why streaming instead of a midnight batch:

- ~22% of emergency admissions arrive between 00:00 and 06:00 AM.
- With a midnight batch, an overnight patient admitted at 01:30 would miss the 07:00 worklist and wait 24+ hours for a case manager.

How it handles hospital bursts:

- Partitioned by (facility_id, encounter_id) so admission and insurance updates stay in exact time order.
- Buffers afternoon peaks (15–30 msg/s at 11:00–16:00) without slowing down the hospital EHR.

SCHEDULED BATCH · APACHE AIRFLOW

1. Nightly Incremental Batch (01:00 AM) — 4. Coding & Billing:

- Pulls newly coded stays (WHERE updated_at > last_run).
- Why not real-time streaming? Coders assign final DRGs 5–14 days after discharge; streaming keystrokes adds database load for zero Day-1 value.

2. Morning API Poll (06:00 AM) — 5. Nursing Home Beds (uhyy-xp9s):

- Pulls open beds at ~85 regional nursing homes right before 06:30 scoring (aligned with nursing-home shift handovers).

3. Yearly Batch — 6. Public NY SPARCS & US Census:

- Runs download_sparcs.py (50,000 rows/page) and logs queries in manifest.json.

THE DAILY TIMELINE · FROM OVERNIGHT ADMISSION TO THE 08:00 HUDDLE

00:00 – 06:00

Overnight Stream

Admissions & ED events land in S3 Bronze (Avro).

→

01:00 & 06:00

Airflow Batches

01:00 coded billing sync; 06:00 fresh SNF bed poll.

→

06:15 – 06:30

Clean, Test & Score

Polars ETL + 4 dbt checks; model flags top 30% (~77–82).

→

07:00 & 08:00

Worklist & Write-Back

07:00 EHR worklist live; 08:00 claim syncs back to lake.

All overnight admissions before 06:00 are scored at 06:30 and ready on the case manager's screen by 07:00.

Row-based Avro for fast stream appends; columnar Parquet for 14× smaller scans

All incoming data are saved unchanged in s3://flhs-bronze/. We use two binary formats and avoid plain CSV or JSON.

APACHE AVRO · ROW-BASED BINARY (FOR STREAMS)

Used for: Kafka streams (HIS/ADT, EDIS, Nursing Note flags).

- Fast stream writes: Events arrive one patient at a time. Because Avro is row-based, Kafka flushes micro-batches straight to disk without reorganising into columns.
- Schema evolution: Stores the schema in the file header so hospital EHR upgrades never break the pipeline.
- **Why not JSON? JSON repeats column names on every row (3–5× larger) and has no strict types.**

APACHE PARQUET (ZSTD) · COLUMNAR BINARY (FOR BATCH & ANALYTICS)

Used for: Batch feeds (Billing, SPARCS, SNF Beds) + all Silver & Gold tables.

- 14× smaller than CSV: Hospital columns repeat values ('Finger Lakes', 'Medicare'). Parquet shrinks our 6,228,309 NY stays in ./data/ from ~1.2 GB CSV to just 83 MB (~13.9 bytes/stay).
- Fast analytical reads: Queries read only needed columns and skip blocks via footer min/max stats.
- **Why not CSV? Loses data types ('120 +' text) and scans every column of every row.**

PARTITIONING ON S3

By source, hospital & date

source=<name>/facility_id=<id>/
year=YYYY/month=MM/day=DD/

- Morning jobs read only today's folder.
- Re-running one hospital never scans the other 16.

15-MIN / 64 MB BUFFER

Avoids the 'small-files problem'

Kafka buffers stream events for 15 min before writing an Avro file.

- 10-sec flushes would create millions of tiny files.
- 1-hr flushes would delay data past the 06:00 cut-off.

7-YEAR RETENTION POLICY

Hot replay + $1/TB/yr archive

- 0–90 Days: S3 Standard ($23/TB/mo) for quick re-runs if an ETL bug is found.
- 90 Days–7 Years: S3 Glacier Deep Archive (~$1/TB/yr) to meet NY medical audit rules.

Real storage measurement: ./WP1/data/B_stays_statewide_2017..2023.parquet = 6,228,309 stays in 83 MB total.

Five rules clean the data and separate Day-1 inputs from outcomes

1 · Unify patient ID across the 17 hospitals (EMPI)

Links visits across Strong Memorial, Highland, etc. to one hashed master_patient_id to track 30-day readmissions.

2 · Deduplicate update messages (ADT^A08)

Keeps the latest message per stay and saves two clean snapshots: what was known at 06:00 on Day 1 and the latest state.

3 · Standardise codes & extract social flags from notes

Maps local insurance names to the 10 SPARCS payers; turns intake notes into lives_alone_flag & prior_nursing_home_flag.

4 · Apply WP1 adult med-surg scope & impute missing values

Applies download_sparcs.py Rules 1–5 (flags deaths/transfers kpi_eligible=False). Fills missing payer with 'Pending' + is_imputed=True instead of dropping patients.

5 · Strict anti-leakage split (Fixed 06:00 AM cut-off)

Splits stays into silver_stay_day1 (inputs known by 06:00 Day 1) and silver_stay_discharged (post-discharge LOS, final DRG, disposition, costs).

BEFORE (RAW BRONZE) → AFTER (CURATED SILVER) · ONE STAY

Field	Before: Raw (Bronze)	After: Curated (Silver)
Stay & ID	local_mrn = 'SMH-8841' msg_id = 'ADT_A08_2'	encounter_id = 'enc_9f3a' master_patient_id = 'pat_4c12'
Payer	' excellus medicaid pending '	payment_typology_1 = 'Medicaid' payer_is_imputed = False
Intake Note	'79yo F, lives alone on 2nd fl, denies prior SNF'	lives_alone_flag = True prior_nursing_home_flag = False
LOS & Ruler	length_of_stay = '120 +' (DRG 194, Severity 3)	los = 120, los_capped = True p75_used = 6 → excess_bed_days = 114

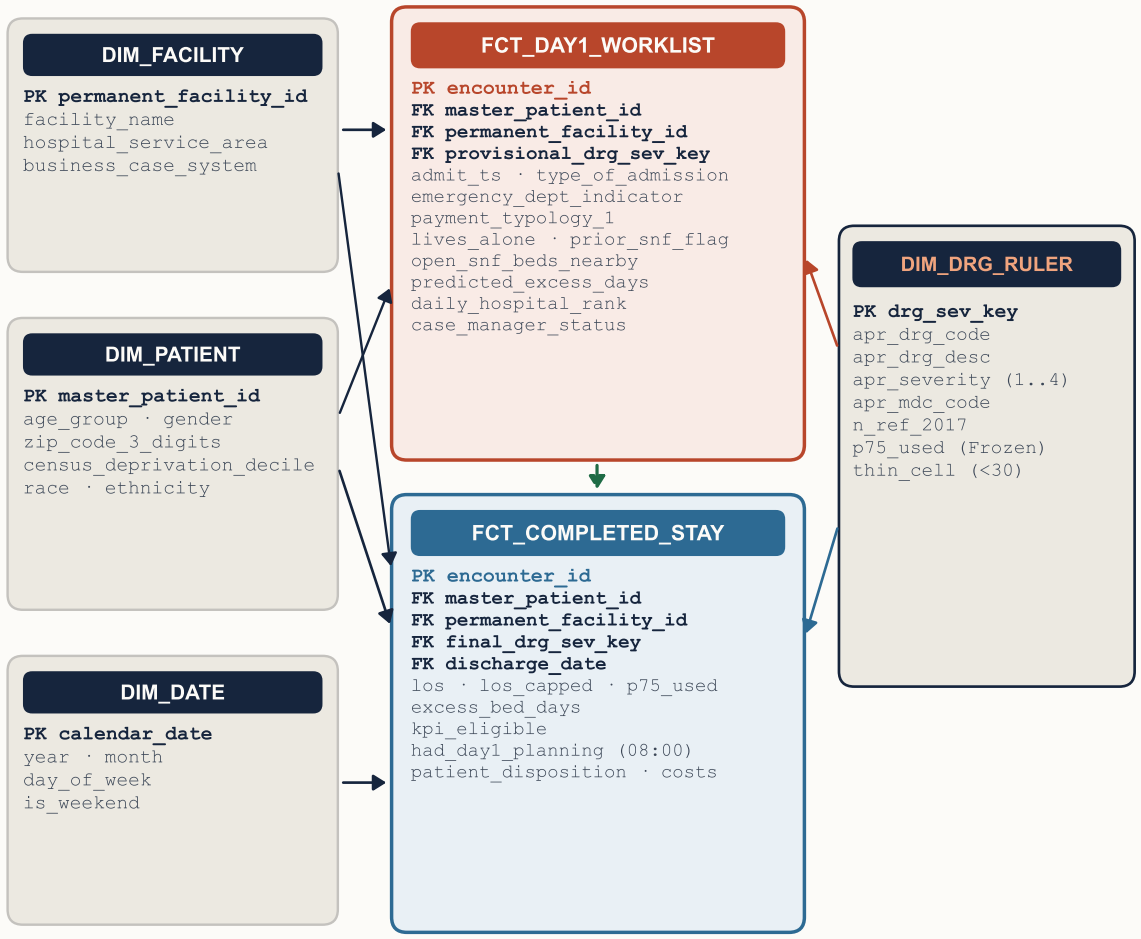
POLARS TRANSFORMATION SNIPPET

```
silver_df = (
  pl.read_parquet('s3://flhs-bronze/stays/*.parquet')
  .sort('event_ts').unique(subset=['encounter_id'], keep='last')
  .with_columns([
    pl.col('payment_typology_1').str.strip_chars().str.to_lowercase()
      .replace(PAYER_MAP, default='Pending').alias('payment_typology_1'),
    pl.col('length_of_stay').str.contains(r'\+').alias('los_capped'),
    pl.col('length_of_stay').str.replace('+','').str.strip_chars()
      .cast(pl.Int32).alias('los'),
  ])
  .join(ruler_2017, on=['apr_drg_code', 'apr_severity_of_illness_code'], how='left')
  .with_columns(
    pl.max_horizontal(0, pl.col('los') - pl.col('p75_used')).alias('excess_bed_days')
  )
)
```

Silver tables use Apache Iceberg on S3, keeping both the 06:00 Day-1 snapshot and the final discharged stay.

One Star Schema across two databases, closed by an 08:00 write-back loop

GOLD STAR SCHEMA (2 FACT TABLES · 4 DIMENSIONS)



ON-PREM POSTGRESQL · FCT_DAY1_WORKLIST

- B-Tree Index on (permanent_facility_id, admit_ts, daily_hospital_rank): Loads each hospital's morning worklist in <15 ms for 45 case managers.
- 06:30 Upsert: INSERT ... ON CONFLICT (encounter_id) DO UPDATE flags the top 30% (~77–82 patients/day).

CLOUD REDSHIFT & ICEBERG · FCT_COMPLETED_STAY

- Partition & Sort Key: Partitioned by discharge_year and sorted by (facility, DRG, severity) to skip irrelevant blocks (Zone Maps).
- Nightly MERGE INTO on encounter_id as billed stays arrive.

CLOSING THE LOOP · 08:00 CASE MANAGER WRITE-BACK

- At 08:00, when a case manager takes a flagged patient on the EHR screen, PostgreSQL records case_manager_status = 'CLAIMED_DAY1'.
- **Syncs back to set had_day1_planning = True in FCT_COMPLETED_STAY—tracking our WP1 Leading Indicator (% of flagged patients started on Day 1).**

DIM_DRG_RULER is frozen on the 2017 NY reference year so the KPI benchmark never drifts.

Every Gold column is tagged by source and Day-1 anti-leakage status

YES · DAY-1 SAFE INPUT | 06:30 SCORE / 08:00 WRITE-BACK | STATIC 2017 RULER | NO · POST-DISCHARGE TARGET ONLY

Column	Table(s)	Type	Source	Known Day 1?	Role & Validation Rule
encounter_id	Both Facts	STRING	HIS / ADT	YES (PK)	Primary key (1 row = 1 stay). NOT NULL, Unique.
master_patient_id	Facts, DIM_PATIENT	STRING	HIS (EMPI)	YES (FK)	Hashed patient ID across 17 hospitals (tracks readmissions).
permanent_facility_id	Facts, DIM_FACILITY	STRING	HIS / SPARCS	YES (FK)	6-digit NY hospital ID (e.g. 000413 Strong Memorial).
admit_ts / discharge_date	Facts, DIM_DATE	TS/DATE	HIS / ADT	Admit: YES	Exact admission hour & discharge date (fixes SPARCS gap).
age_group, gender, zip_3	DIM_PATIENT	STRING	HIS / SPARCS	YES	Patient demographics at admission (age_group != '0 to 17').
census_deprivation_decile	DIM_PATIENT	INT	US Census ADI	YES	Neighbourhood deprivation score (1..10) from 3-digit ZIP.
type_of_admission	FCT_DAY1_WORKLIST	STRING	HIS / ADT	YES	Emergency, Urgent, Elective, Trauma (Newborn excluded).
emergency_dept_indicator	FCT_DAY1_WORKLIST	STRING	ED System	YES	'Y' if admitted through the ED, 'N' otherwise.
payment_typology_1	FCT_DAY1_WORKLIST	STRING	HIS / ADT	YES	Primary payer (Medicare, Medicaid, Private, etc.).
provisional_drg_sev_key	FCT_DAY1_WORKLIST	STRING	ED System	YES (FK)	Working diagnosis & triage severity (1..4) at admission.
lives_alone, prior_snf_flag	FCT_DAY1_WORKLIST	BOOL	EHR Notes	YES	Extracted on-prem from Day-1 nursing intake note.
open_snf_beds_nearby	FCT_DAY1_WORKLIST	INT	SNF Bed API	YES	Open regional nursing-home beds accepting payer at 06:00.
predicted_excess_days, rank	FCT_DAY1_WORKLIST	FLT/INT	06:30 Scorer	06:30 SCORE	Model score & hospital rank (1..N); flags top 30% (~77–82/d).
had_day1_planning	FCT_COMPLETED_STAY	BOOL	08:00 EHR UI	08:00 LOOP	True when case manager claims patient on Day 1 (Leading KPI).
p75_used, thin_cell	DIM_DRG_RULER	INT/BOOL	2017 SPARCS	STATIC RULER	Frozen 2017 NY P75 stay for DRG × severity (fallback n < 30).
los, excess_days, disp, costs	FCT_COMPLETED_STAY	INT/FLT	Billing / Derived	NO (TARGET)	Post-discharge outcomes: MAX(0, los - p75_used). Never Day-1 input.

Note: race & ethnicity in DIM_PATIENT are used only for fairness checks across patient groups—never as model inputs.

7 · SCALABILITY, INFRASTRUCTURE SIZING & FINOPS

Right-sized serverless compute runs all 17 hospitals for ~$750 a month

Measure	Finger Lakes (17 Hospitals)	All NY State (~200 Hospitals)
Total admissions	137,867 / yr (~380/day)	2,125,754 / yr (~5,800/day)
Adult stays scored on Day 1	99,880 / yr (~275/day)	1,471,788 / yr (~4,030/day)
KPI-eligible stays (excl. deaths)	94,132 / yr (~258/day)	1,390,226 / yr (~3,810/day)
Top 30% flagged on worklist	~77–82 patients / day	~1,140–1,210 / day
HL7 stream (avg / peak)	~15k/d (0.18/s · 30/s peak)	~225k/d (2.6/s · 300/s peak)
Daily new data volume	~50 MB / day (~18 GB/yr)	~750 MB / day (~270 GB/yr)
4-yr historical stays (./data/)	434,465 rows (~6 MB Parquet)	6,228,309 rows (83 MB Parquet)

TOTAL CLOUD COST · FINGER LAKES (17 HOSPITALS)

~$600 – $900 / mo

~$10k / yr (< 0.06% of WP1's $17.5M / yr net savings)

S3 + Glacier (~$30/mo) · MSK Serverless (~$40/mo) · Fargate Polars + Airflow (~$180/mo) · Redshift + Glue (~$450/mo).

1 · STREAMING (<0.2 MSG/S AVG, 30 MSG/S PEAK)

MSK Serverless instead of 3 fixed Kafka nodes

- 17 hospitals send only ~15,000 messages/day with afternoon bursts.
- A fixed 3-node cluster ($600+/mo) would sit 99% idle; MSK Serverless handles peaks for ~$40/mo and scales to 300 msg/s statewide.

2 · DAILY 06:30 SCORING (~275 PATIENTS, <50 MB)

Scale UP on 1 Polars container (<10 seconds)

- Scoring 275 patients fits easily in memory on one small container (4 vCPU, 16 GB RAM) running Polars + LightGBM ($0.02/run).
- Avoids paying for a 24/7 Spark cluster that takes minutes just to start.

3 · YEARLY RETRAINING (6.2M – 20M+ ROWS)

Scale OUT on demand with AWS Glue (PySpark)

- Spins up 10 serverless PySpark workers only when retraining on multi-year statewide history.
- Broadcasts DIM_DRG_RULER (16 KB) & DIM_FACILITY (<50 KB) in memory, then shuts down to $0.

Compute sizing follows Session 07 rules: single-node Polars for <10 GB daily jobs; distributed PySpark for multi-year backfills.

Four automated dbt checks and three known production limits

4 AUTOMATED DBT QUALITY CHECKS (RUN AT 06:15 BEFORE SCORING)

1. Uniqueness: encounter_id unique in both Facts; (DRG, severity) unique in DIM_DRG_RULER.
2. Completeness: facility, admit_ts, age, and admission type NOT NULL; alerts IT if any major hospital sends 0 overnight admissions.
3. Validity & Anti-Leakage: severity IN {1..4}; age != '0 to 17'; no feature recorded after 06:00 Day 1.
4. Freshness: MAX(admit_ts) >= 05:30 AM across main ED hubs before 06:30 scoring starts.

DATA CATALOGUE ENTRY · GOLD.FCT_DAY1_WORKLIST

- **Owner: Director of Patient Flow · Data Platform Team**
- Grain: 1 row per adult admission (~275/day; top 30% = ~77–82 flagged)
- Refresh SLA: Daily at 06:30 AM (ready before 07:00 shift start)
- **Privacy Tag: PSEUDONYMISED in Cloud · PHI_RESTRICTED On-Premise**
- Lineage: HIS/ADT + EDIS + Notes + SNF API → silver_stay_day1 → LightGBM
- **Quality Status: 4 / 4 Automated dbt Checks Passing (Freshness < 30 min)**

HOW WP2 SOLVES ALL 6 PUBLIC SPARCS GAPS FROM WP1

Public SPARCS Gap (WP1)	How WP2 Fixes It in Production
1. No patient ID	Linked across 17 hospitals via master_patient_id (EMPI).
2. No dates, only year	Exact admit_ts & discharge_date from HIS/ADT.
3. DRG coded at discharge	Uses ED provisional diagnosis on Day 1.
4. '120 +' cap & hidden IDs	Internal HIS keeps exact LOS & hospital ID.
5. No social factors	Extracted from Day-1 nursing note + Census ZIP score.
6. No bed or SNF census	Live ward events + 06:00 NY SNF bed API poll.

KNOWN LIMITS OF THE NEW SYSTEM (& MITIGATIONS)

1. Provisional vs. final diagnosis drift (~15–20% of complex stays):
- The ED doctor's Day-1 diagnosis can change if complications arise later.
- Mitigation: Re-score in-house patients on Day 3 if a new diagnosis or ICU transfer is logged.

2. Late-night nursing note lag (05:00–06:00 AM arrivals):
- Intake notes for 05:30 arrivals may not be typed by 06:30. Mitigation: Flag as is_imputed = True and score from ED/ADT data first.

3. Regional boundary of patient ID (17 Finger Lakes hospitals):
- Tracks readmissions across the 17 regional hospitals, but cannot see stays outside the region.

Governance and quality checks follow Session 07 (Slides 26 & 92) and Session 08 healthcare standards.